

saasPlug – Canonical Naming & Consistency Sheet

Microservices

1. API Gateway (frontend requests , routing to other services might need auth)
2. Auth Service (Login, sign up for users, sign up for providers, role resolution: χρηστης admin ή provider) → core objects: User, AuthSession, Role
3. User service(ιδία με τλ1: profile , preferences , history etc, vehicles etc) → core objects: UserProfile, Vechicle, PaymentMethod etc
4. Charger service(provider data, find chargers, availability , info etc) → core objects:
5. Reservation service(make and cancel reservations) → core objects: (same as TΛ1)
6. Provider Service(provider registration , provider profile, external api configuration manage owned chargers) → core objects: Provider , ProviderAccount, ProviderApiConfig, SubscriptionPlan and ProviderCharger
7. Billing Service(Provider billing and payment, calculate usage, create invoice, payment status , payment) → core objects: Invoice, InvoiceLine, Payment, BillingPeriod
8. Analytics service(provider analytics, global analytics for operator , usage summaries) → core objects: ProviderAnalyticsReport, GlobalAnalyticsReport, UsageMetric, ExportJob)

1. Actors

- EVUser
- ProviderAdmin
- PlatformOperator
- GoogleOAuthProvider

- ExternalProviderAPI
 - PaymentGateway
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2. Microservices

- ApiGateway
 - AuthService
 - UserService
 - VehicleService
 - ChargerService
 - ReservationService
 - SessionService
 - BillingService
 - ProviderService
 - AnalyticsService
 - IntegrationService
 - AuditService
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3. Service Ownership

Entity / Responsibility	Service
Authentication	AuthService
User profile	UserService
Cars / ownership	VehicleService
Chargers	ChargerService
Availability	ChargerService
Reservations	ReservationService
Sessions	SessionService
User payments	BillingService

Entity / Responsibility	Service
User invoices	BillingService
Provider accounts	ProviderService
Provider API config	ProviderService
Provider billing	BillingService
Analytics	AnalyticsService
Export data	AnalyticsService
Webhooks	IntegrationService
Audit logs	AuditService

4. Persistency Entities

Identity

- User
- Role

Vehicles

- Car
- CarOwnership
- CarColor

Charging

- Charger
- PricingProfile
- WholesalePricePoint
- ChargerStatus
- ConnectorType

Reservation / Sessions

- Reservation
- ReservationStatus
- Session
- SessionStatus

Billing (User-side)

- PaymentMethod
- PaymentAuth
- PaymentStatus
- Invoice

Provider / SaaS

- Provider
- ProviderAccount
- ProviderApiConfig
- ProviderSubscription
- ProviderPlan
- ProviderInvoice
- ProviderPayment
- ProviderUsageRecord

Analytics

- ProviderAnalyticsReport
- GlobalAnalyticsReport
- ExportJob

Integration / Audit

- WebhookSubscription

- WebhookEvent
 - AuditEvent
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5. Invoice Naming

- Invoice → EV user charging invoice
 - ProviderInvoice → SaaS billing invoice
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6. Enums

- Role
- CarColor
- ChargerStatus
- ConnectorType
- ReservationStatus
- SessionStatus
- PaymentStatus

Provider-related

- ProviderStatus
 - SubscriptionStatus
 - ProviderInvoiceStatus
 - ProviderPaymentStatus
 - ExportStatus
 - ExportFormat
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7. DTO Naming (Class – Data Struct)

Pattern

- `<Entity>Dto`
- `<Entity>SummaryDto`
- `<Entity>DetailsDto`

Examples

- UserDto
- CarDto
- CarOwnershipDto
- ChargerDto
- ChargerSummaryDto
- ChargerDetailsDto
- ReservationDto
- ReservationSummaryDto
- ReservationDetailsDto
- SessionDto
- InvoiceDto
- ProviderDto
- ProviderInvoiceDto
- ProviderAnalyticsReportDto
- GlobalAnalyticsReportDto
- ExportJobDto
- WebhookEventDto
- AuditEventDto

8. API Naming (Class – APIs)

Auth

- GoogleLoginRequest
- AuthTokenResponse

Charger

- SearchChargersRequest
- SearchChargersResponse
- GetChargerDetailsResponse

Reservation

- CreateReservationRequest
- CreateReservationResponse
- CancelReservationRequest
- ReservationResponse

Session

- StartSessionRequest
- StartSessionResponse
- StopSessionRequest
- StopSessionResponse

Billing (User)

- AddPaymentMethodRequest
- PaymentMethodResponse
- PaymentAuthorizationResponse
- InvoiceResponse

Provider

- RegisterProviderRequest
- RegisterProviderResponse

- ProviderProfileResponse

Provider Billing

- GetProviderInvoiceResponse
- PayProviderInvoiceRequest
- PayProviderInvoiceResponse

Analytics

- GetProviderAnalyticsRequest
 - ProviderAnalyticsResponse
 - GetGlobalAnalyticsRequest
 - GlobalAnalyticsResponse
 - ExportUsageDataRequest
 - ExportUsageDataResponse
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9. API Classes

- AuthController
 - UserController
 - VehicleController
 - ChargerController
 - ReservationController
 - SessionController
 - BillingController
 - ProviderController
 - AnalyticsController
 - WebhookController
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10. Relationships

- User owns CarOwnership
 - CarOwnership references Car
 - User creates Reservation
 - Reservation targets Charger
 - Reservation may create Session
 - User starts Session
 - Session occurs at Charger
 - Session has PaymentAuth
 - Session generates Invoice
 - User owns PaymentMethod
 - Provider manages Charger
 - Provider has ProviderApiConfig
 - Provider receives ProviderInvoice
 - ProviderInvoice is settled by ProviderPayment
 - AnalyticsService generates ProviderAnalyticsReport
 - PlatformOperator views GlobalAnalyticsReport
 - Provider requests ExportJob
 - WebhookSubscription receives WebhookEvent
 - AuditEvent references User or Session
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11. Use Cases

EVUser

- LoginWithGoogle
- SearchChargers

- ViewChargerDetails
- CreateReservation
- CancelReservation
- StartSession
- StopSession

ProviderAdmin

- RegisterProvider
- ConfigureProviderAPI
- ViewProviderAnalytics
- ViewProviderInvoice
- PayProviderInvoice
- ExportUsageData

PlatformOperator

- ViewGlobalAnalytics
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12. Naming Conventions

- Use exact names from this document
- No synonyms (Reservation, Charger, Invoice)
- PascalCase for classes
- camelCase for fields
- Same names across all diagrams